Final Project Proposal Checkpoint 1

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1. I plan to explore global carbon dioxide emissions and explore how emissions vary across countries and over time. I aim to analyze trends in CO_2 emissions and examine how factors might have contributed to such high emissions. This might include energy consumption and economic output among countries. Maybe the lower the economy a nation has, the less carbon emission footprint they would have compared to higher-wealth countries.

2. I wanted to do something related to climate change or environmental science because it's one of the global issues I care about. I believe every scientist when they talk about how rising greenhouse gas emissions are ruining the planet and there's so much scientific evidence to back that up as well. This was one of the "complex" challenges I hoped to solve or at least tell a meaningful story with just being passionate about data science and engineering. I believe data can play an important role in helping people understand the scale and urgency of climate change. Maybe in the future, we can reverse it but by analyzing emissions data, I hope to tell meaningful stories with the data and better understand the patterns behind global pollution. I hope that it will help people become more aware as we as a community try to mitigate those effects and care about Earth a little more.

3. This problem relates to the course because it's a high-level challenge that is still being researched today. It's a real scientific question that leads to data-driven insight and requires real data and extensive data cleaning and preparation. Examining relationships, modeling, interpreting, and communicating it are all very important aspects that is what makes data science the way it is.

4. First, like every data scientist, I plan to preprocess and clean the data by selecting relevant variables. If it's too many countries, I might focus on a specific region like North America. I will then do EDA to explore trends. I also want to implement unsupervised learning as do PCA or UMAP to explore patterns in the data. I also want to do supervised learning like random forest or any model to show how variables relate to CO_2 emissions and better evaluate the relative importance of these predictors to see which factors are most strongly associated with emission levels.

5. https://ourworldindata.org/co2-and-greenhouse-gas-emissions